BISWAJIT PATTANAIK

Embedded Systems & IoT Engineer
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Summary

Electronics and Communication Engineering undergraduate with hands-on experience in embedded firmware, PCB design, IoT systems, and Edge AI. Proficient in C/C++, Embedded Linux, and FreeRTOS, designing hardware-software solutions on ESP32 and STM32 platforms.

Education

B.Tech in Electronics and Communication Engineering | **CUTM, Bhubaneswar** 2023 – Present
Centurion University of Technology and Management | CGPA: 8.20/10.00

Experience

Intern, Signal & Telecommunication (S&T) | **Indian Railways – East Coast Railway (ECoR)** Jun 2026 – Jul 2026

- Hands-on exposure to networking, data communication, Linux system operations, wireless communication, and RailNet railway communication infrastructure; studied how S&T systems and AI support railway operations.
- Co-developed **"Bharatiya Rail – Track Fault Detection"**, an AI-assisted inspection tool with image-upload and live-camera fault detection, automatic location/time capture, inspector details entry, an AI chatbot, and automated report generation.

Intern, PLC Programming and Embedded C | **CTTC, Bhubaneswar** May 2025 – Jun 2025

- Engineered PLC-based industrial automation programs using Ladder Logic, streamlining process control across three simulated production workflows.
- Programmed STM32 and Arduino microcontrollers in Embedded C, optimizing hardware interfacing routines and reducing memory usage by approximately 20%.
- Configured and tested sensors/actuators within control systems, achieving sub-100 ms response times.

Projects

AI-Powered Vehicle Black Box System | github.com/YOUR-GITHUB-USERNAME/vehicle-black-box 2025
Tech: Raspberry Pi, Python, TensorFlow Lite, OpenCV, Deep Learning, Computer Vision, Flask, GPS, GSM, MQ-3, DHT22

- **Problem:** Accident analysis and emergency response need on-vehicle telemetry and driver monitoring without server round trips.
- **Design:** Pi node (GPS, GSM, MQ-3, DHT22, camera) running an on-device TensorFlow Lite 5-class classifier (accident, dangerous road, drowsiness, etc.) trained on 9,992 images; 99.40% training and 99.25% validation accuracy; 60% lower latency than server-side processing.
- **Built:** Flask dashboard with live GPS tracking and GSM emergency alerts within two seconds of anomaly detection.

Axion – 6-Axis Robot Arm Controller | github.com/YOUR-GITHUB-USERNAME/axion-robot-arm
Tech: KiCad, STM32F103C8T6, I2C, CAN (TCAN330), CP2102N, NRF24L01, SWD, TPS5331x

- **Problem:** Driving a 6-axis arm needs coordinated servo control, per-joint orientation feedback, and clean power on one compact, open board.
- **Design:** STM32F103C8T6 core with six servo outputs, each behind its own MOSFET driver stage; six I2C headers for per-joint IMUs; 12V input stepped to 5V (TPS5331x buck) then 3.3V (LM3480).
- **Built:** Full KiCad PCB with USB (CP2102N), CAN bus (TCAN330), NRF24L01 wireless socket, and SWD debug header, released as open hardware.

Smart Fall Detection and Emergency Response System | github.com/YOUR-GITHUB-USERNAME/smart-fall-detection
Tech: ESP32, ESP32-CAM, TensorFlow Lite, Wi-Fi/HTTP, REST API, HTML/CSS/JS

- **Problem:** Falls must be detected and answered within seconds, even when no one is nearby.
- **Design:** ESP32-CAM runs a TensorFlow Lite fall model; an ESP32 controller reads ultrasonic, PIR, sound, proximity, and pulse sensors over Wi-Fi/HTTP.
- **Built:** Automated buzzer and motor-driven response, plus a live web dashboard with REST endpoints for instant alerts and sensor streaming.

Technical Skills

- **Embedded Platforms:** STM32, ESP32, Raspberry Pi, Arduino, FreeRTOS, Embedded Linux
- **Languages:** Embedded C, C++, Python, Bash
- **Protocols:** UART, SPI, I2C, CAN, BLE, MQTT, GSM, PWM
- **Hardware Design:** PCB Design, Schematic Design, Sensor Interfacing, Power Supply Design, SWD Debugging
- **Edge AI & IoT:** TensorFlow Lite, OpenCV, Flask, REST APIs
- **Software Tools:** KiCad, Proteus, PlatformIO, Git, Linux

Certifications & Achievements

- **Winner – IEEE Hackathon TechSpark 2.0**, for the AI-Powered Vehicle Black Box System (2025)
- **NPTEL Elite** – Introduction to Internet of Things, IIT Kharagpur (2025)
- **PLC Programming** – Industrial Automation (2025)